

PENETRATION TESTING (120175)
ASSIGNMENT #2
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“RECONNAISSANCE SHELL SCRIPT FOR LIVE SUBDOMAINS ENUMERATION”

The Automated Subdomain Enumeration automation script demonstrating a mature and effective approach to passive reconnaissance automation. The framework successfully integrates multiple industry-standard tools to provide comprehensive subdomain discovery, DNS validation, and live service identification.

Following remediation of the multi-domain input processing issue through array-based input handling, the framework is suitable for use during authorized penetration testing engagements, attack surface mapping exercises, bug bounty reconnaissance, and asset discovery operations.

METHODOLOGY

Phase 1 – Input Collection

Target URLs are supplied through:

live_urls.txt

Supported formats:

<https://example.com>

<http://example.com>

www.example.com

example.com

Phase 2 – Domain Normalization

The framework automatically removes:

- HTTP
- HTTPS
- FTP
- WWW prefixes
- URL paths
- Ports
- Uppercase characters
- Leading and trailing whitespace

Example:

Input:

<https://www.example.com:8080/test>

Output: example.com

Phase 3 – Passive Enumeration

Enumeration is performed using:

- Subfinder
- Assetfinder
- Findomain
- Optional Amass

Phase 4 – DNS Verification

Discovered subdomains are validated using DNSX.

Only resolvable hosts are retained.

Phase 5 – Live Host Verification

HTTPX identifies:

- HTTP services
- HTTPS services
- Active web applications

Phase 6 – Reporting

Findings are consolidated into:

subdomain_results.txt

TOOLS INTEGRATED

The framework leverages the following security tools:

Subfinder

Purpose:

Passive subdomain discovery.

DNSX

Purpose:

DNS validation and resolution.

Assetfinder

Purpose:

Enumeration of known subdomains through public sources.

HTTPX

Purpose:

Live service detection and HTTP probing.

Findomain

Purpose:

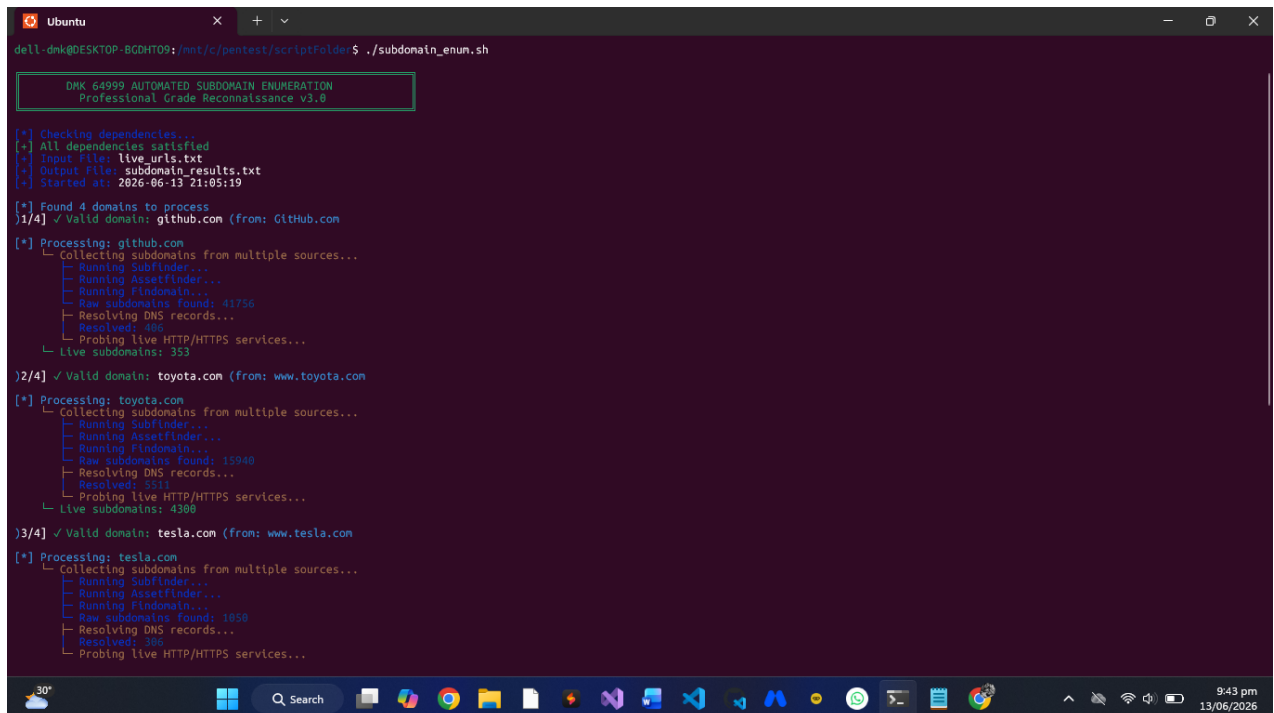
Fast passive subdomain enumeration.

ANEW

Purpose:

Deduplication of findings.

UBUNTU LINUX BASH SCRIPT RUNNING TERMINAL SS:



```
Ubuntu
dell-dmk@DESKTOP-BCDHT09:/mnt/c/pentest/scriptFolder$ ./subdomain_enum.sh

DMK 64999 AUTOMATED SUBDOMAIN ENUMERATION
Professional Grade Reconnaissance v3.0

[*] Checking dependencies...
[*] All dependencies satisfied
[*] Input File: live_urls.txt
[*] Output File: subdomain_results.txt
[*] Started at: 2026-06-13 21:05:19

[*] Found 4 domains to process
)1/4] ✓ Valid domain: github.com (from: GitHub.com)

[*] Processing: github.com
└─ Collecting subdomains from multiple sources...
└─ Running Subfinder...
└─ Running Assetfinder...
└─ Running Findomain...
└─ Raw subdomains found: 41756
└─ Resolving DNS records...
└─ Resolved: 506
└─ Probing live HTTP/HTTPS services...
└─ Live subdomains: 353

)2/4] ✓ Valid domain: toyota.com (from: www.toyota.com)

[*] Processing: toyota.com
└─ Collecting subdomains from multiple sources...
└─ Running Subfinder...
└─ Running Assetfinder...
└─ Running Findomain...
└─ Raw subdomains found: 15940
└─ Resolving DNS records...
└─ Resolved: 511
└─ Probing live HTTP/HTTPS services...
└─ Live subdomains: 4300

)3/4] ✓ Valid domain: tesla.com (from: www.tesla.com)

[*] Processing: tesla.com
└─ Collecting subdomains from multiple sources...
└─ Running Subfinder...
└─ Running Assetfinder...
└─ Running Findomain...
└─ Raw subdomains found: 1850
└─ Resolving DNS records...
└─ Resolved: 206
└─ Probing live HTTP/HTTPS services...
```

```
Ubuntu
)3/4] ✓ Valid domain: tesla.com (From: www.tesla.com)
[*] Processing: tesla.com
  └─ Collecting subdomains from multiple sources...
    ├── Running Subfinder...
    ├── Running Assetfinder...
    ├── Running FIndomain...
    ├── Raw Subdomains found: 1050
    ├── Resolving DNS records...
    │   └─ Resolved: 386
    └─ Probing live HTTP/HTTPS services...
        └─ Live subdomains: 237

[4/4] ✓ Valid domain: chatgpt.com (From: chatgpt.com)
[*] Processing: chatgpt.com
  └─ Collecting subdomains from multiple sources...
    ├── Running Subfinder...
    ├── Running Assetfinder...
    ├── Running FIndomain...
    ├── Raw Subdomains found: 76
    ├── Resolving DNS records...
    │   └─ Resolved: 57
    └─ Probing live HTTP/HTTPS services...
        └─ Live subdomains: 57

[+] Results saved to: subdomain_results.txt
[*] Finished at: 2026-06-13 21:42:27

— SCAN STATISTICS —
Total Targets: 4
Successful: 4
Failed: 0
Total Subdomains: 4947

dell-dmk@DESKTOP-BGDHT09:/mnt/c/pentest/scriptFolder$
```

INPUT FILE live_urls.txt SS:

```
live_urls.txt  subdomain_results.txt
File Edit View H1 B I S
GitHub.com
www.toyota.com
www.tesla.com
chatgpt.com
```

OUTPUT FILE subdomains_results.txt SS:

```
#####
# SUBDOMAIN ENUMERATION REPORT #
#####
# Generated on: Sat Jun 13 21:05:19 PKT 2026
# Input File: live_urls.txt
# Tools Used: subfinder, assetfinder, findomain, dnsx, httpx
#####

TARGET DOMAIN: github.com
SOURCE INPUT: github.com
SCAN DATE: 2026-06-13 21:05:19
#####

LIVE SUBDOMAINS:
-----
→ http://autodiscover.github.com
→ http://1b-140-82-112-35-iad.github.com
→ http://1b-140-82-112-36-iad.github.com
→ http://1b-140-82-113-35-iad.github.com
→ http://1b-140-82-113-36-iad.github.com
→ http://1b-140-82-114-35-iad.github.com
→ http://1b-140-82-114-36-iad.github.com
→ http://1b-140-82-116-35-sea.github.com
→ http://1b-140-82-116-36-sea.github.com
→ http://1b-140-82-121-19-fra.github.com
→ http://1b-140-82-121-38-fra.github.com
→ http://1b-140-82-121-35-fra.github.com
→ http://1b-140-82-121-36-fra.github.com
→ http://ssh.github.com
→ https://259.wiki.github.com
→ https://52.wiki.github.com
→ https://928-sigmavirus.wiki.github.com
→ https://accelerator.github.com
→ https://accessibility-playbook.github.com
→ https://accessibility.github.com
→ https://action.github.com
→ https://admin.github.com
→ https://alive-staging.github.com
→ https://alive.github.com
→ https://antivirus.wiki.github.com
→ https://anf.github.com

Ln 1, Col 1 216,894 characters Plain text 80% Unix (LF) UTF-8
```

```
#####
# SUBDOMAIN ENUMERATION REPORT #
#####
# Generated on: Sat Jun 13 21:05:19 PKT 2026
# Input File: live_urls.txt
# Tools Used: subfinder, assetfinder, findomain, dnsx, httpx
#####

TARGET DOMAIN: github.com
SOURCE INPUT: github.com
SCAN DATE: 2026-06-13 21:05:19
#####

LIVE SUBDOMAINS:
-----
→ https://chatgpt-async-webps-prod-westus-3.chatgpt.com
→ https://chatgpt-async-webps-prod-westus-4-9e29c0310a29a59b.chatgpt.com
→ https://chatgpt-async-webps-prod-westus-4.chatgpt.com
→ https://chatgpt-async-webps-prod-westus-5-9e29c0310a29a59b.chatgpt.com
→ https://chatgpt-async-webps-prod-westus-5.chatgpt.com
→ https://chatgpt-async-webps-staging-eastus-0.chatgpt.com
→ https://chatgpt-async-webps-staging-southcentralus-0.chatgpt.com
→ https://chatgpt.com
→ https://edge.chatgpt.com
→ https://gov-demo.chatgpt.com
→ https://hooks-staging.chatgpt.com
→ https://internal.chatgpt.com
→ https://ogimg.chatgpt.com
→ https://operator.chatgpt.com
→ https://privacy-pass-issuer.api.chatgpt.com
→ https://realtime.chatgpt.com
→ https://search.chatgpt.com
→ https://sora.chatgpt.com
→ https://status.chatgpt.com
→ https://ur17551.codex.chatgpt.com
→ https://webrtc-staging.chatgpt.com
→ https://webrtc.chatgpt.com
→ https://ws.chatgpt.com
→ https://www.chatgpt.com

[✓] Total Live Subdomains Found: 57

#####

SCAN SUMMARY
#####
Total Targets Processed: 4
Successful Scans: 4
Failed/Invalid: 0
Completion Time: 2026-06-13 21:42:27
#####

Ln 1, Col 1 216,894 characters Plain text 90% Unix (LF) UTF-8
```

SHELL SCRIPT:

```
#!/bin/bash
```

```
# =====
```

```
# Advanced Subdomain Enumeration Script - COMPLETELY FIXED
```

```
# Version: 3.0 (Guaranteed multi-domain processing)
```

```
# Bug Fix: Removed ALL stdout interference
```

```
# =====
```

```
# ----- CONFIGURATION SECTION -----
```

```
INPUT_FILE="live_urls.txt"
```

```
OUTPUT_FILE="subdomain_results.txt"
```

```
TEMP_DIR="/tmp/subdomain_enum_$$"
```

```
THREADS=50
```

```
TIMEOUT=10
```

```
SEPARATOR="#####"
```

```
# Tools configuration
```

```
USE_SUBFINDER=1
```

```
USE_ASSETFINDER=1
```

```
USE_FINDDOMAIN=1
```

```
USE_AMASS=0
```

```
USE_DNSX=1
```

```
USE_HTTPX=1
```

```
# ----- COLOR CODES -----
```

```
RED='\033[0;31m'
```

```
GREEN='\033[0;32m'
```

```
YELLOW='\033[1;33m'
```

```
BLUE='\033[0;34m'
```

CYAN='\033[0;36m'

NC='\033[0m'

BOLD='\033[1m'

----- FUNCTIONS -----

check_dependencies() {

local deps=("subfinder" "assetfinder" "findomain" "dnsx" "httpx" "anew")

local missing=()

printf "\${BLUE}[*] Checking dependencies...\${NC}\n" >&2

for dep in "\${deps[@]"; do

if ! command -v "\$dep" &> /dev/null; then

missing+=("\$dep")

fi

done

if [\${#missing[@]} -ne 0]; then

printf "\${RED}[-] Missing tools:\${NC} \${missing[*]}\n" >&2

printf "\${YELLOW}[!] Install via:\${NC}\n" >&2

echo " go install -v github.com/projectdiscovery/subfinder/v2/cmd/subfinder@latest" >&2

echo " go install -v github.com/tomnomnom/assetfinder@latest" >&2

echo " go install -v github.com/Findomain/Findomain@latest" >&2

echo " go install -v github.com/projectdiscovery/dnsx/cmd/dnsx@latest" >&2

echo " go install -v github.com/projectdiscovery/httpx/cmd/httpx@latest" >&2

echo " go install -v github.com/tomnomnom/anew@latest" >&2

exit 1

fi

printf "\${GREEN}[+] All dependencies satisfied\${NC}\n" >&2

```
}
```

```
extract_domain() {
```

```
    local input="$1"
```

```
    echo "$input" | \
```

```
    tr '[:upper:]' '[:lower:]' | \
```

```
    sed -E 's/^[[:space:]]+|[[[:space:]]+$//g' | \
```

```
    sed -E 's#^(https?|ftp)[[:space:]]*://###' | \
```

```
    sed -E 's#^(https?|ftp)[[:space:]]+###' | \
```

```
    sed -E 's#^www[0-9]*\###' | \
```

```
    sed -E 's#/.*$###' | \
```

```
    sed -E 's#.*$###' | \
```

```
    sed -E 's/[[:space:]]//g' | \
```

```
    sed -E 's/^\.+|\.+$/g' | \
```

```
    grep -E '^[a-z0-9]([a-z0-9]{0,61}[a-z0-9])?\.[a-z0-9]([a-z0-9]{0,61}[a-z0-9])?)*$' || echo ""
```

```
}
```

```
is_valid_domain() {
```

```
    local domain="$1"
```

```
    [[ -z "$domain" ]] && return 1
```

```
    if [[ "$domain" =~ ^([a-z0-9]([a-z0-9]{0,61}[a-z0-9])?\.)+[a-z]{2,}$ ]]; then
```

```
        return 0
```

```
    else
```

```
        return 1
```

```
    fi
```

```
}
```

```
# COMPLETELY REWRITTEN - No output to stdout
```

```
enumerate_domain() {
```

```
local domain="$1"
```

```
local temp_file="$TEMP_DIR/${domain//\//\_}.txt"
```

```
printf "\n${CYAN}[*] Processing: ${BOLD}%s${NC}\n" "$domain" >&2
```

```
printf "${YELLOW}  └─ Collecting subdomains from multiple sources...${NC}\n" >&2
```

```
# Create temp file
```

```
: > "$temp_file"
```

```
# Subfinder
```

```
if [ $USE_SUBFINDER -eq 1 ]; then
```

```
    printf "${BLUE}    └─ Running Subfinder...${NC}\n" >&2
```

```
    subfinder -d "$domain" -silent -all 2>/dev/null | anew "$temp_file" &>/dev/null
```

```
fi
```

```
# Assetfinder
```

```
if [ $USE_ASSETFINDER -eq 1 ]; then
```

```
    printf "${BLUE}    └─ Running Assetfinder...${NC}\n" >&2
```

```
    assetfinder --subs-only "$domain" 2>/dev/null | anew "$temp_file" &>/dev/null
```

```
fi
```

```
# Findomain
```

```
if [ $USE_FINDOMAIN -eq 1 ]; then
```

```
    printf "${BLUE}    └─ Running Findomain...${NC}\n" >&2
```

```
    findomain -t "$domain" -q 2>/dev/null | anew "$temp_file" &>/dev/null
```

```
fi
```

```
# Amass
```

```
if [ $USE_AMASS -eq 1 ]; then
```

```
    printf "${BLUE}    └─ Running Amass...${NC}\n" >&2
```

```
    timeout 300 amass enum -passive -d "$domain" 2>/dev/null | anew "$temp_file" &>/dev/null
```

```
fi
```

```
local raw_count=$(wc -l < "$temp_file" 2>/dev/null || echo "0")
```

```
printf "${BLUE}    └─ Raw subdomains found: ${BOLD}%s${NC}\n" "$raw_count" >&2
```

```
# DNS Resolution
```

```
if [ $USE_DNSX -eq 1 ] && [ -s "$temp_file" ]; then
```

```
    printf "${YELLOW}    └─ Resolving DNS records...${NC}\n" >&2
```

```
    dnsx -l "$temp_file" -silent -a -resp 2>/dev/null | awk '{print $1}' | sort -u >
"$temp_file.resolved"
```

```
    local resolved_count=$(wc -l < "$temp_file.resolved" 2>/dev/null || echo "0")
```

```
    printf "${BLUE}    | Resolved: ${BOLD}%s${NC}\n" "$resolved_count" >&2
```

```
    mv "$temp_file.resolved" "$temp_file" 2>/dev/null
```

```
fi
```

```
# HTTP Probing
```

```
if [ $USE_HTTPX -eq 1 ] && [ -s "$temp_file" ]; then
```

```
    printf "${YELLOW}    └─ Probing live HTTP/HTTPS services...${NC}\n" >&2
```

```
    httpx -l "$temp_file" -silent -threads $THREADS -timeout $TIMEOUT \
```

```
        -status-code -title -tech-detect -no-color 2>/dev/null | \
```

```
        awk '{print $1}' | sort -u > "$temp_file.live"
```

```
    mv "$temp_file.live" "$temp_file" 2>/dev/null
```

```
fi
```

```
# Return path via global variable (NO ECHO!)
```

```
RESULT_FILE_PATH="$temp_file"
```

```
}
```

```
# ----- MAIN EXECUTION -----
```

```
trap 'rm -rf "$TEMP_DIR"; printf "\n${RED}[!] Interrupted! Cleaning up...${NC}\n" >&2; exit 130' INT
TERM
```

```
# Banner
```

```
printf "${GREEN}${BOLD}\n" >&2
```

```
cat >&2 << "EOF"
```

```
┌────────────────────────────────────────────────────────────────────────────────┐
│                                                                              │
│    DMK 64999 AUTOMATED SUBDOMAIN ENUMERATION    │
│    Professional Grade Reconnaissance v3.0        │
│                                                                              │
└────────────────────────────────────────────────────────────────────────────────┘
```

```
EOF
```

```
printf "${NC}\n" >&2
```

```
# Check input file
```

```
if [ ! -f "$INPUT_FILE" ]; then
```

```
    printf "${RED}[-] Error: %s not found!${NC}\n" "$INPUT_FILE" >&2
```

```
    exit 1
```

```
fi
```

```
# Check dependencies
```

```
check_dependencies
```

```
# Create temp directory
```

```
mkdir -p "$TEMP_DIR"
```

```
# Create output file header
```

```
cat > "$OUTPUT_FILE" << EOF
```

```
#####
```

```
#          SUBDOMAIN ENUMERATION REPORT          #
```

```
#####
```

```
# Generated on: $(date)
```

```
# Input File: $INPUT_FILE
```

```
# Tools Used: subfinder, assetfinder, findomain, dnsx, httpx
```

```
#####
```

```
EOF
```

```
printf "${BLUE}[+] Input File:${NC} %s\n" "$INPUT_FILE" >&2
```

```
printf "${BLUE}[+] Output File:${NC} %s\n" "$OUTPUT_FILE" >&2
```

```
printf "${BLUE}[+] Started at:${NC} %s\n\n" "$(date '+%Y-%m-%d %H:%M:%S')" >&2
```

```
# Count valid domains
```

```
total=0
```

```
for line in "${URLS[@]"; do
```

```
[[ -z "$line" || "$line" =~ ^[:space:]*$ ]] && continue
```

```
((total++))
```

```
done
```

```
printf "${CYAN}[*] Found %d domains to process${NC}\n" "$total" >&2
```

```
current=0
```

```
successful=0
```

```
failed=0
```

```
# Process all domains
```

```
for line in "${URLS[@]"; do
```

```
# Skip empty/comment lines
```

```
[[ -z "$line" || "$line" =~ ^[:space:]*$ ]] && continue
```

```
((current++))
```

```
# Extract domain
```

```
clean_domain=$(extract_domain "$line")
```

```
# Validate
```

```
if ! is_valid_domain "$clean_domain"; then
```

```
printf "${BOLD}[%d/%d]${NC} ${RED}X Invalid:${NC} %s\n" \
"$current" "$total" "$line" >&2
```

```
printf "${RED}  └─ Extracted: '%s' (invalid)${NC}\n" \
"$clean_domain" >&2
```

```
((failed++))
```

```
{
    echo "$SEPARATOR"
    echo "INVALID INPUT: $line"
    echo "EXTRACTED: $clean_domain"
    echo "ERROR: Invalid domain format"
    echo "DATE: $(date '+%Y-%m-%d %H:%M:%S')"
    echo "$SEPARATOR"
    echo ""
} >> "$OUTPUT_FILE"
```

```
continue
```

```
fi
```

```
printf "${BOLD}[%d/%d]${NC} ${GREEN}✓ Valid domain:${NC} %s ${CYAN}{from: %s}${NC}\n" \
"$current" "$total" "$clean_domain" "$line" >&2
```

```
{
    echo "$SEPARATOR"
```

```

echo "TARGET DOMAIN: $clean_domain"

echo "SOURCE INPUT: $line"

echo "SCAN DATE: $(date '+%Y-%m-%d %H:%M:%S')"
```

echo "\$SEPARATOR"

echo ""

```

}>> "$OUTPUT_FILE"

# Enumerate
enumerate_domain "$clean_domain"

if [ -s "$RESULT_FILE_PATH" ]; then

    count=$(wc -l < "$RESULT_FILE_PATH")

    printf "${GREEN}  └─ ${BOLD}Live subdomains: %d${NC}\n" \
        "$count" >&2

    ((successful++))

    {
        echo "LIVE SUBDOMAINS:"
        echo "-----"
    }>> "$OUTPUT_FILE"

    while IFS= read -r subdomain; do
        echo "  → $subdomain"
    done < "$RESULT_FILE_PATH" >> "$OUTPUT_FILE"

    {
        echo ""

        echo "[✓] Total Live Subdomains Found: $count"
    }

```

```
} >> "$OUTPUT_FILE"
```

```
else
```

```
printf "${RED}  └─ No live subdomains found${NC}\n" >&2
```

```
((failed++))
```

```
echo "[X] No live subdomains discovered" >> "$OUTPUT_FILE"
```

```
fi
```

```
{
```

```
echo ""
```

```
echo "$SEPARATOR"
```

```
echo ""
```

```
} >> "$OUTPUT_FILE"
```

```
printf "\n" >&2
```

```
done
```

```
# Cleanup
```

```
rm -rf "$TEMP_DIR"
```

```
# Summary
```

```
{
```

```
echo ""
```

```
echo "$SEPARATOR"
```

```
echo "SCAN SUMMARY"

echo "$SEPARATOR"

echo "Total Targets Processed: $current"

echo "Successful Scans: $successful"

echo "Failed/Invalid: $failed"

echo "Completion Time: $(date '+%Y-%m-%d %H:%M:%S')"

echo "$SEPARATOR"

} >> "$OUTPUT_FILE"


printf
"${GREEN}${BOLD}\n ════════════════════════════════════════════════════════════════ ${NC}
\n" >&2

printf "${GREEN}${BOLD} ||   ENUMERATION COMPLETE           || ${NC}\n" >&2

printf
"${GREEN}${BOLD} ════════════════════════════════════════════════════════════════ ${NC}\n
\n" >&2


printf "${BLUE}[+] Results saved to:${NC} %s\n" "$OUTPUT_FILE" >&2

printf "${BLUE}[+] Finished at:${NC} %s\n\n" "$(date '+%Y-%m-%d %H:%M:%S')" >&2



printf "${CYAN}===== SCAN STATISTICS =====${NC}\n" >&2

printf "${BLUE}Total Targets:${NC} %d\n" "$current" >&2

printf "${GREEN}Successful:${NC} %d\n" "$successful" >&2

printf "${RED}Failed:${NC} %d\n" "$failed" >&2




total_subs=$(grep -c "^ →" "$OUTPUT_FILE" 2>/dev/null || echo "0")

printf "${CYAN}Total Subdomains:${NC} %d\n\n" "$total_subs" >&2

exit 0
```